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Unit-2, Artificial intelligence notes for AKTU by brevilearning YT compressed

BTech CSE (Dr. A.P.J. Abdul Kalam Technical University)

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Artificial Intelligence

UNIT - 2

(One - Shot)

Topics to be covered:

1. Steps involved in problem solving.
2. Searching
3. Types of search
4. Alpha - Beta Pruning
5. Constraint satisfaction problem.

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Steps involved in problem solving:

Step 1: (Define the problem)

Step 2: (Analyze the problem)

Step 3: (Develop potential solutions)

Step 4: (Evaluate the options)

Step 5: (Select the best option)

Step 6: (Implement the solution)

Step 7: (Measure the results.)

Searching:

- It is a fundamental concept in AI, used in various applications such as route planning, puzzle - solving, optimization problems and decision making systems.
- It refers to the process of systematically exploring a problem space to find a solution.

Key components of searching:

i) State space

ii) Initial state

iii) Goal state

iv) Operators or actions

v) Path

Types of search:

- Informed vs Uninformed search
- DFS vs BFS
- Heuristic search
- Hill climb search and A^* search.

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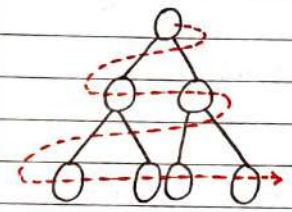
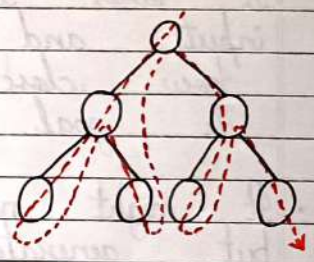
* Informed search	Uninformed search
• Uses knowledge to find steps to solution.	• No use of knowledge.
• Quick solution	• Slow solution
• Low cost	• high cost
• More efficient	• less efficient
• Low time & space complexity	• more time and space complexity
• e.g: Best First Search, A* search.	• e.g: BFS, DFS, etc.

* BFS	DFS
• Breadth first search performs breadth wise searching.	• Depth first search performs depth wise searching.
• Uninformed approaches to traverse tree.	• Uninformed approaches to traverse tree.
• No backtracking is allowed.	• Backtracking is allowed.
• Provides solution in min-steps, if there exist more than one solution.	• Reoccurence of nodes result solution in max-steps.
• More optimal	• Less optimal
• More time consuming	• Faster but can stuck in (∞) loop.
• 	• 

* Best first search:

- Best First Search is combination of Breadth first search and Depth first search.
- It allows switching b/w BFS and DFS, and choose most promising node to go to the goal node.
- If a node is less promising, it jumps over next node in same level.

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Heuristic search:

- Works on heuristic function that ranks alternatives in search algorithm.
- Heuristic function rank the branches based on available information to decide which branch to follow.
- It takes current state of agent as input and compute estimation of how close an agent is from its goal.
- It might not give best solution everytime, but generate guarantee for optimal solution.
- It finally calculates the cost of path with most optimal solution.

#note: Value of HF is always (+ve)

Hill climbing:

- It is based on loop that moves in direction of goal.
- The loop is terminated when it reaches peak.
- It don't maintain search tree, it only record states and functions.

Algorithm:

Step - 1: Evaluate initial state.

Step - 2: if (goal found)
return success;
else
continue;

Step - 3: Step 2 is repeated in a loop until either of these conditions are satisfied.

i) Found the solution
ii) All operations are performed

Step - 4: If (solution not found)
go to step-1;

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ch :

based on informed search technique
heuristic function to reach the
possible solution.

optimal and complete algorithm
ind the solution.

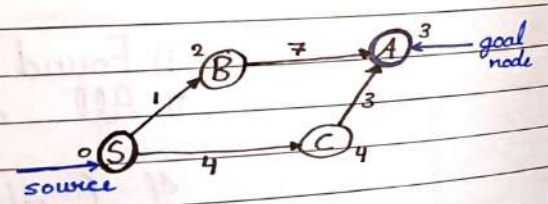
is comparitively faster than any
searching algorithms to
complex problem.

for total cost to reach a
node is given as:

$$f(n) = g(n) + h(n)$$

$h(n)$ → heuristic value (child)

$g(n)$ → cost to reach child node.



given:
S → 0
A → 3
B → 2
C → 4

S → B 3 ✓
S → C 8

S → B → A 11 4

"Subscribe"

Alpha - Beta pruning:

- It is an optimization technique for min-max algorithm, commonly used in game-playing scenarios like chess.
- The goal is to minimize the no. of nodes evaluated in the game tree by eliminating the branches that cannot affect the final decision.
- Min-max algorithm: This is a decision making algorithm used in two-player games. It explores all possible moves, creating a tree of possible game states.
- Alpha (α):
  - represent the best/maximum value that the maximizing player can acheive.
  - initial value : $(-\infty)$
- Beta (β):
  - represents the best/minimum value that the minimizing player can acheive.
  - initial value : (∞)

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- Pruning mechanism:

  - If maximizing player discovers a move of (value ≥ beta), it prunes the remaining branches.

  - If minimizing player finds a move of (value ≤ alpha), it prunes the remaining branches.

Constraint Satisfaction problem:

- Constraint satisfaction problem is a logical approach used to represent a problem in terms of a set of variables, domains and constraints.

- The goal is to assign values to the variables that satisfies all constraints.

key components of CSP:

i) Variables: • They represent unknown decision variables in the problem.

  • Each variable has a domain, which is set of values it can take.

ii) Domains: • Define the possible values that a variable can take.

  • They are usually determined based on the nature of the problem.

iii) Constraints: Defines the relationship or condition that must be satisfied for a solution to be valid.

e.g: Sudoku is best example of CSP;

↳ variables are cells on board,
↳ domain are digits {1-9} to be filled in variables.
↳ constraints are rules of Sudoku.

(such as each row, column and 3X3 subgrid must contain distinct values)

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Artificial Intelligence

UNIT - 2

Topics to be covered

1. Best First Search
2. Numerical on Constraint Satisfaction Problem

Best First Search:

- It is a graph search algorithm that selects the most promising node for expansion based on heuristic function.
- It priortizes nodes based on heuristic evaluation, also the effectiveness of this algorithm is dependent on the choice and accuracy of heuristic function.

Step 1: Initialization

- Start with an initial node as the current state.

Step 2: Heuristic Function

- Define a heuristic function 'h(n)' to estimate the cost or distance from current node to the goal node.

#note: most promising node has minimum value of heuristic function.

Step 3: Validation / Selection

- Select the node with lowest heuristic value and check whether it is goal node or not.
- If not, expand selected node by generating its successor.

Step 4: Repetition

- For successors in expansion, repeat step 2 and 3 for each successor.

Step 5: Termination

- Terminate after finding goal node or after completing no. of iterations.

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Numerical

1. Evaluate constraint satisfaction problem within crypt arithematic problem:

CROSS + ROADS = DANGER

⇒

i) C = ~~8~~ 0 – 9
R = ~~4~~
O =
S = ~~0~~ ~~2~~
A = ~~0~~
D = ~~1~~
N =
G =
E = ~~3~~

e.g:

```
  ONE
+ ONE
  TWO
```

O	N	E	T	W
2	3	1	4	6

```
  2 3 1
  2 3 1
  -----
  4 6 2
```

⇒ {O:2, N:3, E:1, T:4, W:6}


```
    TO
+   NO
  -----
  OUT
```

T = 2
O = 1
N = 8
U = 0

```
  2 1
  8 1
  -----
1 0 2
```

⇒ {T:2, O:1, N:8, U:0}

	C	R	O	S	S
+	R	O	A	D	S
D	A	N	G	E	R

	~~8~~	4		2	2
+	4		0	1	2
1	0			3	4

	9	6	2	3	3
+	6	2	5	1	3
1	5	8	7	4	6

C : 9
R : 6
O : 2
S : 3
A : 5
D : 1
N : 8
G : 7
E : 4

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Let's Help Each Other

No amount is small

Notes are free, it's your choice to help me for my efforts!

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